Technical Report: Apertus Trust-VC (Sovereign Due Diligence Agent)

Event: Hack Apertus 2026 (Track 2B: Own Project)

Team: Parian (solo)

1. Executive Summary & Problem Statement

European Venture Capital (VC) firms, private equity funds, and financial institutions handle highly confidential startup pitch decks, cap tables, and financial statements daily. Under strict European and Swiss data sovereignty laws (GDPR) and corporate security mandates, financial analysts **cannot** transmit sensitive financial data to US-based, proprietary cloud AI infrastructure (such as OpenAI or Google).

Apertus Trust-VC bridges this gap by introducing a sovereign, localized Due Diligence Agent. Designed to run natively on Switzerland's open Large Language Model (**Apertus 70B**), the platform enables secure financial parsing, automated risk assessment, and core business evaluation without any data crossing European or Swiss borders.

2. System Architecture & Tech Stack

The platform is built using a modular, decoupled architecture prioritizing reliability, ease of auditing, and strict data isolation:

- **Frontend & Interface:** Streamlit (providing a fast, reactive web interface for secure PDF uploads and real-time report generation).
- **Agent Orchestration & Prompt Layer:** LangChain & LangChain-Core (managing the execution chain, context windows, and structured prompt engineering).
- **Document Parsing & Extraction:** pypdf (extracting raw text cleanly from multi-page PDF pitch decks and financial reports).
- **Inference Pipeline:** I wanted to design by Hugging Face Inference Endpoints connecting directly to **Apertus 8B/70B** hosted on Swiss supercomputing infrastructure (CSCS). But as it was provided just now and it's late to redo everything, *I remained Temporary proxy LLM endpoints utilized during final integration stages due to endpoint allocation schedules unchanged.*

3. Core Functional Workflows

1. **Secure Document Ingestion:** Users upload institutional financial reports or pitch decks via a localized web portal.
2. **Deterministic Text Extraction:** The extraction engine parses text block-by-block while sanitizing formatting anomalies.
3. **Structured Financial Prompting:** The text context is passed through an expert VC analyst prompt template designed to isolate three critical institutional metrics:

4.

    - *Financial Risks & Liabilities* (Leverage ratios, debt concentration, cash-flow pressures).
    - *Core Business Strengths* (Net margins, operating efficiency, liquidity ratios).
    - *Critical Red Flags* (Hidden reporting inconsistencies, high-risk dividend policies).

5. **Structured Output Generation:** Results are rendered cleanly in professional markdown bullet points for immediate investment committee review.

4. Visual Overview & Architecture Reference

(Embedded visual representation of system workflows and roadmap)

- **Core Interface:** Web-based document upload and automated analysis trigger.
- **Data Flow:** Secure PDF → Text Extraction → LangChain Pipeline → Localized LLM Inference → Structured VC Insights.

5. Security, Compliance & Data Sovereignty

- **Zero Cross-Border Leakage:** Architecture ensures data remains within compliant European/Swiss jurisdictions.
- **Open-Weight Transparency:** Built entirely on open-source frameworks (Python, LangChain, Apertus), allowing complete enterprise auditing and self-hosted on-premise deployment.

6. Future Roadmap & Scaling

- **Fine-Tuning for Financial NLP:** Fine-tuning Apertus 8B/70B specifically on Swiss/EU financial reporting standards and compliance guidelines.
- **Multi-Document Agentic Memory:** Expanding LangChain agent loops to cross-reference multiple historical balance sheets and market reports simultaneously.
- **Enterprise On-Premise Packaging:** Providing Dockerized deployment kits for institutional investors requiring air-gapped internal server setups.
- **UI/UX development**: no successful digital product can skip this crucial step.
- **Human-Centered Product Design:** Venture requires aligning with hymen centered design principles to work the best for its high prestigious and rigorous clients.

0:00 - 0:15

Apertus Trust-VC

Sovereign Due Diligence for European Venture Capital.

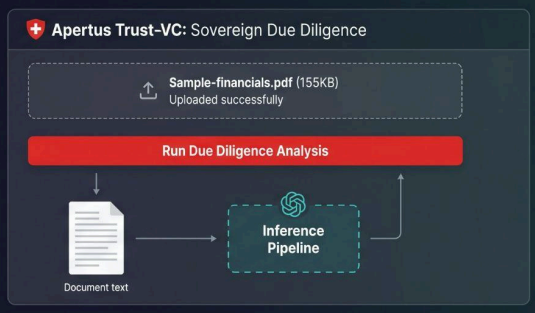


The Privacy Crisis: VC firms and financial institutions cannot send highly confidential pitch decks and financial reports to US-based AI APIs (OpenAI/Google) due to strict EU/Swiss data sovereignty (GDPR).

0:15 - 0:40

Live Demo: Seamless Analysis

Extracting critical venture insights in seconds from any financial PDF.



0:40 - 1:00

Key Venture Insights Extracted

1. Financial Risks & Liabilities

- Long-term debt concentration (61% of liabilities)
- Leverage risks (Debt-to-equity ~2.1x)
- Cash flow tight for debt servicing

2. Core Business Strengths

- 9.8% Net margin ($945K on $9.6M revenue)
- Strong short-term liquidity (Current ratio ~2.6x)
- Retained earnings growth ($820K to $1.265M)

3. Critical Red Flags

- High leverage profile ($8M notes payable)
- Large dividend payouts ($500K) vs. cash generation
- Potential reporting duplication in statements

1:00 - 1:15

The Sovereign AI Backbone

Leveraging Switzerland's Fully Open LLM

- **Built for Security:** Runs entirely on Apertus 70B hosted on Swiss-based servers (CSCS)
- **Data Sovereignty:** Guaranteeing no data leakage outside European and Swiss borders
- **Integration Ready:** Fully compatible with EPFL and ETH Zurich's open-weight framework



1:15 - 1:30

Future Roadmap & Scaling

Where we go from here:

- **Fine-Tuning for Financial NLP:** Customize Apertus 8B/70B specifically on Swiss and European VC compliance datasets.
- **Agentic Workflows:** Integrate LangChain memory and tools for multi-document comparisons (historical performance analysis).
- **Hybrid On-Prem Deployment:** Provide private cloud and localized hosting tailored for risk-averse institutional investors.